

CO4 - AT1

INDUSTRIAL PROBLEM SOLVING

Assessment Report

Question 2 — Online Banking Interface

A GUI-Based Banking Application Developed Using Java Swing

Name	
Register Number	
Degree & Branch	B.Tech – Artificial Intelligence and Data Science
Year / Semester	
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1. Objective of the Application

The objective of this application is to design and implement a **GUI-based online banking interface** that allows a customer to log in securely and perform core banking operations — **balance inquiry, fund transfer, and viewing transaction history** — through a graphical interface built entirely with Java Swing.

The industrial problem being solved is that of **reliable, self-service banking at the customer-facing layer**. In a real bank, an incorrect account number, a negative transfer amount, or an overdrawn account are not simply cosmetic problems; they cause failed settlements, reconciliation errors, and financial loss. This application therefore focuses not only on *performing* transactions but on **preventing invalid transactions from ever entering the system**, using layered input validation, PIN authorisation, and clear error reporting through `JOptionPane`.

The specific objectives are:

1. To build a responsive, professional-looking desktop interface using Java Swing components.
2. To implement secure input handling for sensitive fields — account number, amount, and PIN.
3. To enforce banking business rules such as positive amounts, sufficient balance, minimum balance, and per-transaction limits.
4. To maintain an automatic, tamper-proof (read-only) transaction history using an `ArrayList` displayed through a `JTable`.
5. To demonstrate how validation and event-driven design improve the reliability of a transactional system.

2. Features Implemented

#	Feature	Description
1	Secure Login	Account number + masked PIN, with a maximum of 3 attempts before the session is terminated
2	Balance Inquiry	Displays holder name, account number, available balance, and transferable amount
3	Fund Transfer	Transfers money between two accounts with PIN re-authorisation and a confirmation dialog
4	Transaction History	Automatically updated <code>JTable</code> showing every transaction for the logged-in account
5	Dual-Entry Recording	Every transfer creates a DEBIT record for the sender and a CREDIT record for the receiver
6	Failed-Transaction Audit	Wrong PIN and insufficient-balance attempts are logged with status <code>FAILED</code>
7	Digit-Only Input Filter	Account number and PIN fields physically reject non-numeric keystrokes

#	Feature	Description
8	Minimum Balance Rule	A balance of Rs. 500 must remain after every transfer
9	Per-Transaction Limit	A single transfer cannot exceed Rs. 50,000
10	Self-Transfer Prevention	The user's own account is excluded from the beneficiary list
11	Colour-Coded History	Credits shown in green, debits in dark red, failures in red
12	Live Session Clock	A <code>javax.swing.Timer</code> updates the header every second
13	Status Bar	Shows the outcome of the most recent operation
14	Clear Form / Logout	Resets input fields and destroys the session data on screen
15	Sortable History	Column headers can be clicked to sort records

3. Swing Components Used

Component	Where it is used	Purpose
<code>JFrame</code>	Main application window	Top-level container for the whole application
<code>JPanel</code>	Login panel, header, transfer form, button bar, status bar	Groups related components and applies layouts
<code>JLabel</code>	Field captions, welcome text, balance display, clock, status	Displays non-editable text
<code>TextField</code>	Account number, transfer amount	Accepts visible textual input
<code>JPasswordField</code>	Login PIN, transaction PIN	Accepts masked input; returns <code>char[]</code> instead of <code>String</code>
<code>JButton</code>	Login, Clear, Balance Inquiry, Fund Transfer, Transaction History, Clear Form, Logout	Triggers all user actions
<code>JTable</code>	Transaction history	Displays tabular transaction records
<code>DefaultTableModel</code>	Backing model of the table	Holds rows; overridden to be read-only
<code>JScrollPane</code>	Wraps the <code>JTable</code>	Provides scrolling when records exceed the visible area
<code>JComboBox<Account></code>	Beneficiary selection	Restricts the destination to valid existing accounts

Component	Where it is used	Purpose
<code>JOptionPane</code>	All messages	Success, error, warning, and confirmation dialogs
<code>JTableHeader</code>	Table header styling	Applies the bank's colour theme
<code>DefaultTableCellRenderer</code>	Row rendering	Colour-codes credits, debits, and failures
<code>TitledBorder</code> / <code>EmptyBorder</code>	Section framing and padding	Visual organisation
<code>javax.swing.Timer</code>	Header clock	Periodic UI updates on the Event Dispatch Thread
<code>DocumentFilter</code> (<code>PlainDocument</code>)	Account and PIN fields	Blocks invalid characters as they are typed

4. Event Listeners Used

The application uses the **delegation event model**. The main class itself implements `ActionListener`, so one central `actionPerformed()` method receives every action event and dispatches it by comparing the event source.

```
public class OnlineBankingApp extends JFrame implements ActionListener {
    @Override
    public void actionPerformed(ActionEvent e) {
        Object src = e.getSource();
        if (src == btnLogin)      handleLogin();
        else if (src == btnBalance) handleBalanceInquiry();
        else if (src == btnTransfer) handleFundTransfer();
        else if (src == btnHistory) handleTransactionHistory();
        else if (src == btnClearForm) clearTransferForm();
        else if (src == btnLogout)  handleLogout();
    }
}
```

Listener / Mechanism	Registered on	Event handled
<code>ActionListener</code>	All seven <code>JButton</code> objects	Button clicks
<code>ActionListener</code>	<code>txtLoginAccount</code> , <code>pwdLoginPin</code>	Pressing Enter inside a field submits the login
<code>ActionListener</code> (lambda)	<code>javax.swing.Timer</code>	Fires once per second to refresh the clock
<code>DocumentFilter</code>	Account number and PIN fields	Intercepts every insert/replace before it reaches the document
<code>WindowListener</code> (built-in)	<code>JFrame</code> via <code>setDefaultCloseOperation</code>	Closing the window exits the application

Two listener styles are demonstrated deliberately: the **explicit interface implementation** (`implements ActionListener`), which is the classic textbook approach, and a **lambda expression** for the timer, which is the modern Java 8+ approach.

5. Input Validation

Validation is applied in **three layers**, so an invalid value has to pass through several independent checks before it can reach the account balance.

Layer 1 — Prevention at the keyboard (`DocumentFilter`)

The account number and PIN fields are attached to a custom `DigitOnlyFilter`. A non-digit key or an over-length entry is rejected before it appears on screen, accompanied by a beep.

```
for (int i = 0; i < text.length(); i++) {
    if (!Character.isDigit(text.charAt(i))) {
        Toolkit.getDefaultToolkit().beep();
        return; // keystroke discarded
    }
}
```

Layer 2 — Format validation (regular expressions)

Field	Rule	Regular expression
Account Number	Exactly 6 digits	<code>\d{6}</code>
PIN	Exactly 4 digits	length check on <code>char[]</code>
Amount	Positive number, at most 2 decimals	<code>\d{1,7}(\.\d{1,2})?</code>

Layer 3 — Business-rule validation

The fund transfer performs ten sequential checks before any balance is modified:

Step	Check	Message type
1	A beneficiary is selected	Warning
2	Beneficiary is not the user's own account	Error
3	Amount field is not empty	Warning
4	Amount matches the numeric pattern	Error
5	Amount parses successfully (<code>NumberFormatException</code> guarded)	Error
6	Amount is strictly greater than zero	Error
7	Amount does not exceed the Rs. 50,000 per-transaction limit	Warning
8	The transaction PIN is correct	Error + audit log

Step	Check	Message type
9	Balance after transfer stays above the Rs. 500 minimum	Error + audit log
10	The user confirms the transfer in a <code>YES_NO</code> dialog	Confirmation

Only when all ten checks pass is the debit/credit pair executed. This is the **fail-fast** principle: the state of the accounts is never touched until every rule has been satisfied.

6. Error Handling

All errors are reported through `JOptionPane` with a message type chosen to match the severity:

Situation	Dialog type	Message shown
Empty account number or PIN	<code>WARNING_MESSAGE</code>	Fields cannot be empty
Account number not 6 digits	<code>ERROR_MESSAGE</code>	Invalid account number
PIN not 4 digits	<code>ERROR_MESSAGE</code>	Invalid PIN
Unknown account or wrong PIN	<code>ERROR_MESSAGE</code>	Invalid credentials + attempts remaining
3 failed logins	<code>ERROR_MESSAGE</code>	Account locked, application exits
No beneficiary selected	<code>WARNING_MESSAGE</code>	Please select a beneficiary
Transfer to own account	<code>ERROR_MESSAGE</code>	Invalid beneficiary
Empty amount	<code>WARNING_MESSAGE</code>	Amount required
Non-numeric / negative amount	<code>ERROR_MESSAGE</code>	Invalid amount, with the offending text echoed
Amount above Rs. 50,000	<code>WARNING_MESSAGE</code>	Limit exceeded
Wrong transaction PIN	<code>ERROR_MESSAGE</code>	Authorisation failed
Insufficient balance	<code>ERROR_MESSAGE</code>	Available, minimum, maximum transferable, and requested amounts
Successful transfer	<code>INFORMATION_MESSAGE</code>	Txn ID, amount, beneficiary, timestamp, new balance
Confirmation before transfer	<code>QUESTION_MESSAGE</code> (Yes/No)	Full summary of the pending transfer

Two defensive programming techniques are also used:

- `try...finally` around login and transfer, so the PIN character array is wiped and the password field cleared on *every* exit path — normal return, error return, or exception.
- `try...catch (NumberFormatException)` around `Double.parseDouble()`, so even if the regular expression were bypassed the application reports a friendly message instead of crashing with a stack trace.

An important design decision: the login failure message deliberately says *"Invalid Account Number or PIN"* rather than identifying which one was wrong. Revealing which field was correct would tell an attacker that a particular account number exists — a real-world vulnerability known as **user enumeration**.

7. Layout Managers Used and Why

Layout Manager	Where used	Why it was chosen
<code>CardLayout</code>	Root container	Swaps between the Login screen and the Dashboard as two complete "cards", giving a clean session boundary — the dashboard simply does not exist on screen until login succeeds
<code>BorderLayout</code>	Login panel, dashboard, header, south section	The natural choice for a screen with a fixed banner (NORTH), an expanding work area (CENTER), and a fixed action bar (SOUTH); the CENTER region absorbs all extra space when the window is resized
<code>GridBagLayout</code>	Login form, fund transfer form	The only standard layout that aligns labels and fields of differing sizes on a common baseline, and the <code>weightx</code> property lets the amount and PIN fields grow proportionally when the window widens
<code>GridLayout</code>	Action button bar, banner text, header right block	Forces all five action buttons to exactly equal width and height, giving the toolbar a professional, uniform appearance
<code>FlowLayout</code>	Login button row, status bar	Ideal for a small number of components that should sit next to each other with natural spacing and wrap if needed

Nesting these layouts (a `GridLayout` panel placed inside the SOUTH region of a `BorderLayout`, which itself sits inside a `CardLayout`) is what makes the interface **responsive**: no coordinate is ever hard-coded, so the window can be resized or maximised and every component repositions itself correctly.

8. How Fund Transfer Works

The transfer is implemented as a **two-legged, all-or-nothing operation**.

Step 1 — Collect input. The beneficiary comes from a `JComboBox` pre-populated with real `Account` objects (the user's own account is excluded), the amount from a `JTextField`, and the authorisation PIN from a `JPasswordField`.

Step 2 — Validate. The ten checks listed in Section 5 run in sequence. Any failure produces a `JOptionPane` message and returns immediately, leaving both balances untouched.

Step 3 — Confirm. A `JOptionPane.showConfirmDialog` displays the sender, beneficiary, and amount. The user can still cancel here.

Step 4 — Commit. Only after confirmation are the balances modified:

```
currentUser.debit(amount);    // money leaves the sender
beneficiary.credit(amount);   // money arrives at the receiver

String txnId = "TXN" + (++txnCounter);
String now   = dateFormat.format(new Date());

transactions.add(new Transaction(currentUser.getAccountNumber(), txnId, now, "DEBIT",
    beneficiary.getAccountNumber(), amount, currentUser.getBalance(), "SUCCESS"));
transactions.add(new Transaction(beneficiary.getAccountNumber(), txnId, now, "CREDIT",
    currentUser.getAccountNumber(), amount, beneficiary.getBalance(), "SUCCESS"));
```

The debit and credit are written as **two records sharing one transaction ID**. This mirrors the double-entry bookkeeping used by real banks: the sender sees a DEBIT in their history and the receiver sees a matching CREDIT in theirs, and the two can always be reconciled through the shared ID.

Step 5 — Refresh the interface. The header balance, the history table, and the status bar are updated, the form is cleared, and a success dialog reports the transaction ID and the new balance.

Worked example (verified during testing): Arun Kumar begins with Rs. 25,000.00 and transfers Rs. 2,500.50 to Priya Sharma. His balance becomes Rs. 22,499.50 and a DEBIT row appears in his history; Priya's balance rises to Rs. 51,001.00 and a CREDIT row with the same transaction ID appears in hers. A subsequent attempt to transfer Rs. 49,000 is rejected with an insufficient-balance error and recorded as `FAILED`, while the balance remains exactly Rs. 22,499.50.

9. How Transaction History Works

Storage. All records live in a single collection declared in the main class:

```
private final ArrayList<Transaction> transactions = new ArrayList<>();
```

Each `Transaction` object stores the owning account number, transaction ID, date and time, type (`OPENING` / `DEBIT` / `CREDIT`), the counter-party account, the amount, the balance after the transaction, and the status (`SUCCESS` / `FAILED`).

Display. The `JTable` is driven by a `DefaultTableModel` with seven columns. The method `refreshHistoryTable()` clears the model and re-adds only those records whose owner matches the logged-in account:

```
historyModel.setRowCount(0);
for (Transaction t : transactions) {
    if (t.getOwner().equals(currentUser.getAccountNumber())) {
        historyModel.addRow(new Object[]{ t.getTxnId(), t.getDateTime(), t.getType(),
            t.getCounterAccount(), String.format("%.2f", t.getAmount()),
            String.format("%.2f", t.getBalanceAfter()), t.getStatus() });
    }
}
```

Automatic updating. `refreshHistoryTable()` is called immediately after every successful transfer, after every logged failure, and on login — so a new transaction appears in the table without the user pressing any refresh button. The **Transaction History** button additionally displays a summary dialog with the total credited, total debited, and current balance.

Integrity. `isCellEditable()` is overridden to return `false`, so the history is strictly read-only — a user cannot type over a record to hide a transaction. Each account sees only its own records, which is a simple form of data isolation. `setAutoCreateRowSorter(true)` still lets the user sort by any column for convenience.

10. How Secure Input Handling Improves Reliability

Secure input handling and reliability are usually treated as separate concerns, but in a transactional system they are the same concern viewed from two angles. The following table shows what each measure prevents.

Secure input measure	Attack or accident prevented	Reliability benefit
<code>JPasswordField</code> with echo character	Shoulder-surfing of the PIN	Credentials cannot be read off the screen
<code>getPassword()</code> returning <code>char[]</code>	PIN persisting in the String pool	The PIN can be, and is, erased from memory
<code>Arrays.fill(pin, '\0')</code> in a <code>finally</code> block	PIN lingering in RAM after use	Guaranteed cleanup on every exit path
<code>DocumentFilter</code> on numeric fields	Letters, symbols, or paste attacks in an account field	<code>NumberFormatException</code> becomes impossible, so the application cannot crash on bad input
Regular-expression format checks	Malformed account numbers and amounts	Only well-formed data reaches the business logic
<code>JComboBox</code> for the beneficiary	Typing a non-existent account number	A transfer can never be sent to an account that does not exist
Positive-amount and limit checks	Negative or absurd amounts creating money	The system cannot be driven into an impossible state
Balance and minimum-balance checks	Overdraft	Account balances can never become negative
PIN re-entry before transfer	Misuse of an unattended logged-in session	Every movement of money is individually authorised
Three-attempt login limit	Brute-force PIN guessing	Automated attacks are stopped early
Vague login failure message	User enumeration	Attackers learn nothing about which accounts exist
Read-only <code>JTable</code>	Editing or hiding a record	The audit trail is trustworthy

Secure input measure	Attack or accident prevented	Reliability benefit
Confirmation dialog	Accidental double-click transfers	Reduces user error, the most common cause of failed transactions

The underlying principle is **fail-fast validation**: rejecting bad data at the earliest possible moment. Because an invalid value can never reach `Double.parseDouble()` or the balance arithmetic, the application has far fewer states in which it can misbehave. Reliability here is not achieved by handling exceptions after the fact, but by making the invalid states unreachable.

11. System Responsiveness During Multiple Operations

Threading model. The application follows the Swing single-thread rule. The user interface is created inside `SwingUtilities.invokeLater()`, so every component is constructed on the **Event Dispatch Thread (EDT)**:

```
SwingUtilities.invokeLater(() -> new OnlineBankingApp().setVisible(true));
```

All event handlers then run on that same thread, which means the balance can never be read while another handler is halfway through modifying it. Serial execution on the EDT gives this application the same safety that a `synchronized` block would give in a multithreaded design — without any locking cost.

Behaviour under repeated operations. During testing, the following sequence was executed without restarting the application:

1. Login → 2. Balance inquiry → 3. Successful transfer → 4. Insufficient-balance attempt → 5. Invalid-amount attempt → 6. Wrong-PIN attempt → 7. Transaction history → 8. Logout → 9. Login as a different user.

Observations:

- Every operation completed instantly; no perceptible delay was introduced by any handler.
- The header balance, the status bar, and the history table stayed consistent with each other after every operation.
- The failed attempts left the balance completely unchanged, confirming that validation runs before any state mutation.
- After logout the history table was cleared and the beneficiary list emptied, so no data leaked into the next session.
- The session clock continued ticking once per second throughout, which is direct visual proof that the EDT was never blocked.

Why it stays responsive. Each handler performs only a linear scan of small in-memory collections, so its running time is negligible. `JOptionPane` dialogs are modal, which is intentional: while a dialog is open the user cannot start a second, conflicting operation. This serialises user actions in exactly the way a banking terminal should behave.

Where responsiveness would break down. If a real database or network call replaced the in-memory `ArrayList`, that call would run on the EDT and freeze the interface for its entire duration. The correct remedy is `SwingWorker`, which performs the slow work on a background thread and publishes results back to the EDT — this is listed in Section 13 as the first future improvement.

12. Limitations of the Current System

1. **No persistence.** All accounts and transactions live in memory; closing the application discards every transfer and resets the balances to their sample values.
2. **PINs stored in plain text.** The `Account` class holds the PIN as a `String`. A production system would store only a salted hash (for example PBKDF2 or bcrypt) and compare hashes.
3. **Single-user session.** Only one customer can be logged in at a time, and the application is not networked.
4. **No true atomicity.** The debit and credit are two ordinary Java statements. If an error occurred between them, money could be lost. A real system would wrap them in a database transaction with commit/rollback.
5. **No concurrency control.** Because everything runs on the EDT there is no corruption today, but if the application were extended to multiple threads or clients, the balance updates would need `synchronized` blocks or a `ReentrantLock`.
6. **double used for currency.** Floating-point arithmetic can accumulate tiny rounding errors; `BigDecimal` is the correct type for monetary values.
7. **Hard-coded business rules.** The minimum balance, transaction limit, and attempt limit are compile-time constants rather than configurable parameters.
8. **No session timeout.** A logged-in session stays open indefinitely if the user walks away.
9. **Limited history features.** There is no date-range filter, search, pagination, or export to CSV/PDF, so the table would become unwieldy with thousands of records.
10. **No account management.** New accounts cannot be opened, PINs cannot be changed, and there are no other transaction types such as deposit, withdrawal, or bill payment.
11. **No accessibility or localisation.** The interface is English-only with a fixed currency prefix and no screen-reader support.
12. **No unit tests.** Correctness was verified manually rather than through an automated test suite.

13. Future Improvements

Responsiveness

- Move all long-running work (database queries, network calls) into a `SwingWorker` so the EDT is never blocked.
- Add a `JProgressBar` or a busy cursor for operations that take noticeable time.
- Replace the `Timer`-driven full table rebuild with incremental row insertion for very large histories.

Security

- Store **salted password hashes** instead of plain-text PINs.
- Add **two-factor authentication** using a one-time password.
- Introduce a **session timeout** that logs the user out after a period of inactivity.
- Add **rate limiting and account lockout** with a cool-down period rather than terminating the application.
- Encrypt data at rest and use **TLS** for any network communication.
- Maintain a separate **immutable audit log** with the user, timestamp, IP address, and outcome of every attempt.

Scalability

- Replace the `ArrayList` with a relational database (**MySQL/PostgreSQL via JDBC**) using `PreparedStatement` to prevent SQL injection.
- Adopt a **three-tier architecture** — Swing client, business-logic service layer, persistence layer — so the same logic can serve a web or mobile front end.
- Wrap transfers in **ACID database transactions** with commit and rollback.
- Add **pagination and indexed date-range queries** for transaction history.
- Use **connection pooling** and, for multi-user access, optimistic locking on account rows.

Usability

- Add **keyboard mnemonics and tab-order** tuning for fully keyboard-driven operation.
- Provide a **beneficiary management screen** with nicknames and favourites.
- Add **export to PDF/CSV** and a printable statement.
- Introduce **filters and a search box** above the history table.
- Support **multiple languages and locale-aware currency formatting** via `ResourceBundle` and `NumberFormat`.
- Add a **dark theme** and adjustable font sizes for accessibility.
- Show **email or SMS notification** simulation after each transaction.

Engineering Quality

- Refactor into separate classes and packages (`model`, `service`, `ui`) following the **MVC pattern**.
- Add **JUnit tests** for the validation and transfer logic.
- Introduce **logging** via `java.util.logging` or Log4j instead of relying on dialogs alone.

14. Conclusion

This assignment produced a complete, working online banking interface in a single Java file using only the standard JDK. The application successfully implements the three required operations — balance inquiry, fund transfer, and transaction history — inside a professionally organised Swing interface built from

`JFrame`, `JPanel`, `JLabel`, `JTextField`, `JPasswordField`, `JButton`, `JTable`, `JComboBox`, and `JOptionPane`, and arranged with `CardLayout`, `BorderLayout`, `GridBagLayout`, `GridLayout`, and `FlowLayout`.

The most valuable lesson from the exercise concerns the relationship between **input validation and system reliability**. A banking application does not fail because its arithmetic is wrong; it fails because invalid data was allowed in. By filtering keystrokes at the field level, validating formats with regular expressions, enforcing business rules before any state change, and requiring PIN authorisation for each transfer, the application makes entire classes of failure — crashes on non-numeric input, negative transfers, overdrafts, and unauthorised movements of money — structurally impossible rather than merely unlikely.

The event-driven architecture, with all handlers executing serially on the Event Dispatch Thread, kept the interface responsive across repeated operations while guaranteeing that no two operations could interleave and corrupt a balance. The system's limitations — the absence of persistence, hashed credentials, and true transactional atomicity — are precisely the boundaries where a classroom prototype ends and a production banking system begins, and the improvements identified in Section 13 map directly onto the technologies used in industry: JDBC with ACID transactions, `SwingWorker` for background processing, hashed credentials, and a layered MVC architecture.

REQUIREMENT COVERAGE MATRIX

Every requirement stated in the question, and exactly where it is satisfied in the code.

#	Requirement from the question	How it is satisfied	Location in OnlineBankingApp.java
1	Fund transfer	Full ten-step validated transfer with dual-entry recording	<code>handleFundTransfer()</code>
2	Balance inquiry	Dialog with holder, account, balance, transferable amount	<code>handleBalanceInquiry()</code>
3	Viewing transaction history	<code>JTable</code> filtered to the logged-in account + summary dialog	<code>handleTransactionHistory()</code> , <code>refreshHistoryTable()</code>
4	Use Java Swing for the GUI	Entire UI built with <code>javax.swing</code> ; no AWT-only components	Whole file
5	<code>JFrame</code>	Main class extends <code>JFrame</code>	Class declaration
6	<code>JPanel</code>	Login panel, header, transfer form, action bar, status bar	<code>buildLoginPanel()</code> , <code>buildDashboardPanel()</code>
7	<code>JLabel</code>	Captions, welcome, balance, clock, status, hints	Throughout the builders
8	<code>TextField</code>	<code>txtLoginAccount</code> , <code>txtAmount</code>	Section 4 fields
9	<code>PasswordField</code>	<code>pwdLoginPin</code> , <code>pwdTxnPin</code>	Section 4 fields
10	<code>Button</code>	Seven buttons	<code>makeButton()</code> and builders
11	<code>JTable</code>	Transaction history with custom renderer	<code>buildHistoryTable()</code>
12	<code>ComboBox</code>	Beneficiary selection, typed as <code>JComboBox<Account></code>	<code>buildTransferForm()</code>
13	<code>OptionPane</code>	Every success, error, warning, and confirmation message	All handlers
14	Event listeners for all interactions	Class implements <code>ActionListener</code> ; lambda for the timer; <code>DocumentFilter</code> for typing	<code>actionPerformed()</code>
15	User-friendly and responsive interface	Nested resizable layouts, tooltips, colour coding, status bar, Enter-key submission, hand cursor	<code>buildLoginPanel()</code> , <code>buildDashboardPanel()</code>
16	Secure input handling — account number	Digit-only <code>DocumentFilter</code> limited to 6 characters + regex check	<code>DigitOnlyFilter</code> , <code>handleLogin()</code>

#	Requirement from the question	How it is satisfied	Location in <code>OnlineBankingApp.java</code>
17	Secure input handling — amount	Regex <code>\d{1,7}(\.\d{1,2})?</code> + guarded parse + range checks	<code>handleFundTransfer()</code> steps 3–7
18	Secure input handling — password/PIN	<code>JPasswordField</code> , <code>char[]</code> , digit filter, <code>Arrays.fill()</code> wipe in <code>finally</code>	<code>handleLogin()</code> , <code>handleFundTransfer()</code>
19	Validate that account numbers contain valid input	Exactly six digits; non-digits blocked at the keyboard	<code>handleLogin()</code> validation 2
20	Validate that amounts are positive and within balance	Steps 6 and 9 of the transfer	<code>handleFundTransfer()</code>
21	Prevent transfers when balance is insufficient	Balance minus minimum-balance check aborts before any debit	<code>handleFundTransfer()</code> step 8
22	Handle invalid transactions with <code>JOptionPane</code> error messages	Fourteen distinct dialogs with matching message types	Table in Section 6
23	Implement balance inquiry	Dialog plus live header label	<code>handleBalanceInquiry()</code> , <code>updateHeaderBalance()</code>
24	Implement fund transfer between accounts	Debit sender, credit beneficiary, shared transaction ID	<code>handleFundTransfer()</code> step 10
25	Transaction history using <code>JTable</code>	Seven-column read-only sortable table	<code>buildHistoryTable()</code>
26	Each successful transaction appears automatically	<code>refreshHistoryTable()</code> invoked immediately after commit	<code>handleFundTransfer()</code>
27	Suitable layout managers	<code>CardLayout</code> , <code>BorderLayout</code> , <code>GridBagLayout</code> , <code>GridLayout</code> , <code>FlowLayout</code>	See Section 7
28	Appropriate event listeners such as <code>ActionListener</code>	Implemented on the frame; registered on all buttons and text fields	<code>actionPerformed()</code>
29	Demonstrate how secure input handling improves reliability	Full analysis with a prevention/benefit table	Report Section 10
30	Analyse behaviour under multiple operations	Nine-step test sequence with observations	Report Section 11

#	Requirement from the question	How it is satisfied	Location in <code>OnlineBankingApp.java</code>
31	Suggest improvements	Grouped under responsiveness, security, scalability, usability, engineering quality	Report Section 13
32	Complete working application, not pseudocode	Compiles with <code>javac</code> and runs; verified end to end	Entire file
33	Beginner-friendly	Single class, commented sections, simple <code>if-else</code> dispatch, no design patterns beyond MVC-lite	Whole file
34	Single Java file for IntelliJ/Eclipse	One public class with static inner classes	<code>OnlineBankingApp.java</code>
35	Sample account data, no database	Four accounts seeded with opening-balance records	<code>loadSampleAccounts()</code>
36	<code>ArrayList</code> for transaction history	<code>ArrayList<Account></code> and <code>ArrayList<Transaction></code>	Section 3 fields
37	Visually organised and professional GUI	Navy/white theme, titled borders, striped rows, uniform button bar	Colour constants and builders
38	Buttons: Login, Balance Inquiry, Fund Transfer, Transaction History, Logout/Clear	All present, plus Clear on the login screen	<code>buildLoginPanel()</code> , <code>buildDashboardPanel()</code>
39	<code>JOptionPane</code> for success/error/warning	<code>INFORMATION_MESSAGE</code> , <code>ERROR_MESSAGE</code> , <code>WARNING_MESSAGE</code> , <code>QUESTION_MESSAGE</code> all used	All handlers
40	No external libraries	Imports only <code>javax.swing</code> , <code>java.awt</code> , <code>java.text</code> , <code>java.util</code>	Import block

SCREENSHOTS TO CAPTURE FOR THE REPORT

Run the application and capture the following. The suggested inputs reproduce each screen exactly.

#	Screenshot	How to produce it	Caption to write below the image
1	Login screen	Run the program; capture the initial window	<i>Fig. 1 — Login screen with masked PIN entry and demo credentials</i>
2	Invalid account number error	Type <code>12</code> in the account field, any 4-digit PIN, click Login	<i>Fig. 2 — Format validation rejects an account number that is not six digits</i>
3	Invalid credentials / attempts remaining	Enter <code>999999</code> / <code>1111</code> , click Login	<i>Fig. 3 — Failed login showing remaining attempts without revealing which field was wrong</i>
4	Main banking screen (dashboard)	Log in with <code>100001</code> / <code>1234</code>	<i>Fig. 4 — Main dashboard: header, transfer form, history table, and action bar</i>
5	Balance inquiry	Click Balance Inquiry	<i>Fig. 5 — Balance inquiry dialog showing available and transferable balance</i>
6	Fund transfer — form filled	Select <code>100002 - Priya Sharma</code> , amount <code>2500.50</code> , PIN <code>1234</code> (before clicking)	<i>Fig. 6 — Fund transfer form with beneficiary, amount, and PIN authorisation</i>
7	Transfer confirmation dialog	Click Fund Transfer	<i>Fig. 7 — Confirmation dialog listing sender, beneficiary, and amount</i>
8	Successful transfer	Click Yes on the confirmation	<i>Fig. 8 — Success message with transaction ID, timestamp, and updated balance</i>
9	History updated automatically	Observe the dashboard immediately after the success dialog closes	<i>Fig. 9 — The DEBIT record appears in the history table without a manual refresh</i>
10	Invalid amount error	Enter <code>abc</code> (or <code>-500</code>) as the amount, PIN <code>1234</code> , click Fund Transfer	<i>Fig. 10 — Non-numeric amount rejected before any balance is modified</i>
11	Insufficient balance error	Enter <code>49000</code> as the amount, PIN <code>1234</code> , click Fund Transfer	<i>Fig. 11 — Transfer blocked because the balance would fall below the minimum</i>
12	Wrong transaction PIN	Enter <code>100</code> as the amount with PIN <code>9999</code> , click Fund Transfer	<i>Fig. 12 — PIN re-authorisation failure declines the transfer</i>
13	Transaction history dialog	Click Transaction History	<i>Fig. 13 — History summary showing total credited, debited, and current balance</i>
14	Transaction history table with mixed records	Close the dialog and capture the table showing SUCCESS and FAILED rows	<i>Fig. 14 — Colour-coded history: debits in red, credits in green, failed attempts flagged</i>

#	Screenshot	How to produce it	Caption to write below the image
15	Receiver's credit entry	Logout , log in as <code>100002</code> / <code>2345</code> , click Transaction History	<i>Fig. 15 — The matching CREDIT record in the beneficiary's account with the same transaction ID</i>
16	Logout confirmation	Click Logout	<i>Fig. 16 — Logout confirmation dialog that clears all session data</i>

Tip: on Windows use `Win + Shift + S` (Snipping Tool) and paste directly into your report. Capture the whole application window including the title bar, and make sure the dialog and the window behind it are both visible in the error screenshots — this proves the error was raised by the running application.

SAMPLE OUTPUT — SCREENSHOTS

Paste each captured screenshot in the space provided and keep the caption below it.

[PASTE SCREENSHOT HERE]

Fig. 1 — Login screen with masked PIN entry and demo credentials

[PASTE SCREENSHOT HERE]

Fig. 2 — Format validation rejects an account number that is not six digits

[PASTE SCREENSHOT HERE]

Fig. 3 — Failed login showing remaining attempts without revealing which field was wrong

[PASTE SCREENSHOT HERE]

Fig. 4 — Main dashboard: header, transfer form, history table, and action bar

[PASTE SCREENSHOT HERE]

Fig. 5 — Balance inquiry dialog showing available and transferable balance

[PASTE SCREENSHOT HERE]

Fig. 6 — Fund transfer form with beneficiary, amount, and PIN authorisation

[PASTE SCREENSHOT HERE]

Fig. 7 — Confirmation dialog listing sender, beneficiary, and amount

[PASTE SCREENSHOT HERE]

Fig. 8 — Success message with transaction ID, timestamp, and updated balance

[PASTE SCREENSHOT HERE]

Fig. 9 — The DEBIT record appears in the history table without a manual refresh

[PASTE SCREENSHOT HERE]

Fig. 10 — Non-numeric amount rejected before any balance is modified

[PASTE SCREENSHOT HERE]

Fig. 11 — Transfer blocked because the balance would fall below the minimum

[PASTE SCREENSHOT HERE]

Fig. 12 — PIN re-authorisation failure declines the transfer

[PASTE SCREENSHOT HERE]

Fig. 13 — History summary showing total credited, debited, and current balance

[PASTE SCREENSHOT HERE]

Fig. 14 — Colour-coded history: debits in red, credits in green, failed attempts flagged

[PASTE SCREENSHOT HERE]

Fig. 15 — The matching CREDIT record in the beneficiary's account with the same transaction ID

[PASTE SCREENSHOT HERE]

Fig. 16 — Logout confirmation dialog that clears all session data

SOURCE CODE — OnlineBankingApp.java

```

/* =====
 * CO4-AT1 : INDUSTRIAL PROBLEM SOLVING
 * Question 2 : ONLINE BANKING INTERFACE (Java Swing)
 * -----
 * A single-file, database-free GUI banking application that supports:
 * 1. Secure login (Account Number + PIN)
 * 2. Balance Inquiry
 * 3. Fund Transfer between accounts
 * 4. Transaction History in a JTable
 *
 * HOW TO RUN
 *   javac OnlineBankingApp.java
 *   java OnlineBankingApp
 * (In IntelliJ IDEA / Eclipse: create a project, paste this file as
 * OnlineBankingApp.java, and run the main method.)
 *
 * SAMPLE LOGIN CREDENTIALS
 *   Account No : 100001    PIN : 1234    (Arun Kumar)
 *   Account No : 100002    PIN : 2345    (Priya Sharma)
 *   Account No : 100003    PIN : 3456    (Rahul Verma)
 *   Account No : 100004    PIN : 4567    (Sneha Nair)
 *
 * No external libraries are used - only the standard JDK (javax.swing, java.awt).
 * =====*/

import javax.swing.*;
import javax.swing.border.EmptyBorder;
import javax.swing.border.TitledBorder;
import javax.swing.table.DefaultTableCellRenderer;
import javax.swing.table.DefaultTableModel;
import javax.swing.table.JTableHeader;
import javax.swing.text.AttributeSet;
import javax.swing.text.BadLocationException;
import javax.swing.text.DocumentFilter;
import javax.swing.text.PlainDocument;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.text.SimpleDateFormat;
import java.util.ArrayList;
import java.util.Arrays;
import java.util.Date;

public class OnlineBankingApp extends JFrame implements ActionListener {

    /* =====
     * SECTION 1 : DATA MODEL CLASSES
     * =====*/

    /** Represents one bank account. */
    static class Account {
        private final String accountNumber; // 6-digit unique number
        private final String holderName;
        private final String pin; // 4-digit PIN (plain text - demo only)
        private double balance;

        Account(String accountNumber, String holderName, String pin, double balance) {
            this.accountNumber = accountNumber;
            this.holderName = holderName;
            this.pin = pin;
            this.balance = balance;
        }

        String getAccountNumber() { return accountNumber; }
        String getHolderName() { return holderName; }
        String getPin() { return pin; }
        double getBalance() { return balance; }

        void credit(double amount) { balance += amount; }
        void debit(double amount) { balance -= amount; }

        @Override
        public String toString() { return accountNumber + " - " + holderName; }
    }
}

```



```

/** Represents one entry in the transaction history. */
static class Transaction {
    private final String owner;           // account this record belongs to
    private final String txnId;
    private final String dateTime;
    private final String type;           // CREDIT / DEBIT / OPENING
    private final String counterAccount; // the other party
    private final double amount;
    private final double balanceAfter;
    private final String status;        // SUCCESS / FAILED

    Transaction(String owner, String txnId, String dateTime, String type,
                String counterAccount, double amount, double balanceAfter, String status) {
        this.owner      = owner;
        this.txnId      = txnId;
        this.dateTime   = dateTime;
        this.type       = type;
        this.counterAccount = counterAccount;
        this.amount     = amount;
        this.balanceAfter = balanceAfter;
        this.status     = status;
    }

    String getOwner()      { return owner; }
    String getTxnId()      { return txnId; }
    String getDateTime()   { return dateTime; }
    String getType()       { return type; }
    String getCounterAccount() { return counterAccount; }
    double getAmount()     { return amount; }
    double getBalanceAfter() { return balanceAfter; }
    String getStatus()     { return status; }
}

/* =====
 * SECTION 2 : SECURE INPUT HANDLING - DIGIT-ONLY DOCUMENT FILTER
 * Stops invalid characters at the moment of typing instead of
 * discovering them later. This is the first line of defence.
 * =====*/
static class DigitOnlyFilter extends DocumentFilter {
    private final int maxLength;

    DigitOnlyFilter(int maxLength) { this.maxLength = maxLength; }

    @Override
    public void insertString(FilterBypass fb, int offset, String text, AttributeSet attr)
        throws BadLocationException {
        replace(fb, offset, 0, text, attr);
    }

    @Override
    public void replace(FilterBypass fb, int offset, int length, String text, AttributeSet attr)
        throws BadLocationException {
        if (text == null) return;

        // Reject any character that is not a digit
        for (int i = 0; i < text.length(); i++) {
            if (!Character.isDigit(text.charAt(i))) {
                Toolkit.getDefaultToolkit().beep(); // audible feedback
                return;
            }
        }
        // Reject anything that would exceed the fixed field length
        int newLength = fb.getDocument().getLength() - length + text.length();
        if (newLength > maxLength) {
            Toolkit.getDefaultToolkit().beep();
            return;
        }
        super.replace(fb, offset, length, text, attr);
    }
}

/* =====
 * SECTION 3 : CONSTANTS AND APPLICATION STATE
 * =====*/

private static final int ACC_NO_LENGTH = 6;
private static final int PIN_LENGTH = 4;
private static final int MAX_LOGIN_ATTEMPTS = 3;
private static final double MAX_TXN_AMOUNT = 50000.00; // per-transaction cap
private static final double MIN_BALANCE = 500.00; // minimum balance to keep

```

```

private final ArrayList<Account> accounts = new ArrayList<>();
private final ArrayList<Transaction> transactions = new ArrayList<>();

private Account currentUser = null; // null means "nobody is logged in"
private int loginAttempts = 0;
private int txnCounter = 1000; // used to build TXN ids

private final SimpleDateFormat dateFormat = new SimpleDateFormat("dd-MM-yyyy HH:mm:ss");

/* ---- Colour / font theme ---- */
private static final Color NAVY = new Color(23, 42, 84);
private static final Color BLUE = new Color(30, 95, 170);
private static final Color GREEN = new Color(24, 128, 76);
private static final Color RED = new Color(176, 42, 42);
private static final Color GREY = new Color(96, 96, 100);
private static final Color LIGHT = new Color(242, 245, 250);

private static final Font F_TITLE = new Font("Segoe UI", Font.BOLD, 22);
private static final Font F_LABEL = new Font("Segoe UI", Font.PLAIN, 14);
private static final Font F_BOLD = new Font("Segoe UI", Font.BOLD, 14);
private static final Font F_FIELD = new Font("Consolas", Font.PLAIN, 15);

/* =====
 * SECTION 4 : SWING COMPONENTS
 * ===== */

private CardLayout cardLayout;
private JPanel cardPanel; // holds LOGIN and DASHBOARD screens

/* ---- Login screen ---- */
private JTextField txtLoginAccount;
private JPasswordField pwdLoginPin;
private JButton btnLogin, btnClearLogin;
private JLabel lblLoginMessage;

/* ---- Dashboard ---- */
private JLabel lblWelcome, lblHeaderBalance, lblClock, lblStatus;
private JComboBox<Account> cmbToAccount;
private JTextField txtAmount;
private JPasswordField pwdTxnPin;
private JButton btnBalance, btnTransfer, btnHistory, btnClearForm, btnLogout;
private JTable tblHistory;
private DefaultTableModel historyModel;

/* =====
 * SECTION 5 : CONSTRUCTOR - BUILDS THE WHOLE UI
 * ===== */
public OnlineBankingApp() {
    loadSampleAccounts();

    setTitle("Online Banking Interface | CO4-AT1 Industrial Problem Solving");
    setSize(980, 690);
    setMinimumSize(new Dimension(880, 620));
    setLocationRelativeTo(null); // centre on screen
    setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

    cardLayout = new CardLayout();
    cardPanel = new JPanel(cardLayout); // CardLayout swaps whole screens
    cardPanel.add(buildLoginPanel(), "LOGIN");
    cardPanel.add(buildDashboardPanel(), "DASHBOARD");

    add(cardPanel);
    cardLayout.show(cardPanel, "LOGIN");

    startClock(); // proves the UI stays alive
}

/* ----
 * Sample data - the app works immediately, no database required.
 * ---- */
private void loadSampleAccounts() {
    accounts.add(new Account("100001", "Arun Kumar", "1234", 25000.00));
    accounts.add(new Account("100002", "Priya Sharma", "2345", 48500.50));
    accounts.add(new Account("100003", "Rahul Verma", "3456", 12750.75));
    accounts.add(new Account("100004", "Sneha Nair", "4567", 8300.00));

    // Seed each account with an opening-balance entry so the history
    // table is never empty on the very first login.
    String now = dateFormat.format(new Date());
    for (Account a : accounts) {
        transactions.add(new Transaction(a.getAccountNumber(), "TXN" + (++txnCounter),

```

```

        now, "OPENING", "-", a.getBalance(), a.getBalance(), "SUCCESS"));
    }
}

/* =====
 * SECTION 6 : LOGIN SCREEN (GridBagLayout inside BorderLayout)
 * ===== */
private JPanel buildLoginPanel() {
    JPanel root = new JPanel(new BorderLayout());
    root.setBackground(LIGHT);

    /* ---- Banner (NORTH) ---- */
    JPanel banner = new JPanel(new GridLayout(2, 1));
    banner.setBackground(NAVY);
    banner.setBorder(new EmptyBorder(22, 20, 22, 20));

    JLabel lblBank = new JLabel("SECURE NET BANKING", SwingConstants.CENTER);
    lblBank.setFont(new Font("Segoe UI", Font.BOLD, 26));
    lblBank.setForeground(Color.WHITE);

    JLabel lblSub = new JLabel("Customer Login Portal", SwingConstants.CENTER);
    lblSub.setFont(F_LABEL);
    lblSub.setForeground(new Color(180, 200, 230));

    banner.add(lblBank);
    banner.add(lblSub);
    root.add(banner, BorderLayout.NORTH);

    /* ---- Login form (CENTER) - GridBagLayout gives precise alignment ---- */
    JPanel form = new JPanel(new GridBagLayout());
    form.setBackground(LIGHT);

    GridBagConstraints gbc = new GridBagConstraints();
    gbc.insets = new Insets(10, 12, 10, 12);
    gbc.anchor = GridBagConstraints.WEST;

    JLabel lblHead = new JLabel("Sign in to your account");
    lblHead.setFont(F_TITLE);
    lblHead.setForeground(NAVY);
    gbc.gridx = 0; gbc.gridy = 0; gbc.gridwidth = 2;
    gbc.anchor = GridBagConstraints.CENTER;
    form.add(lblHead, gbc);

    gbc.gridwidth = 1;
    gbc.anchor = GridBagConstraints.WEST;

    JLabel lblAcc = new JLabel("Account Number :");
    lblAcc.setFont(F_BOLD);
    gbc.gridx = 0; gbc.gridy = 1;
    form.add(lblAcc, gbc);

    txtLoginAccount = new JTextField(14);
    txtLoginAccount.setFont(F_FIELD);
    applyDigitFilter(txtLoginAccount, ACC_NO_LENGTH); // secure input handling
    txtLoginAccount.setToolTipText("Enter your 6-digit account number");
    gbc.gridx = 1; gbc.gridy = 1;
    form.add(txtLoginAccount, gbc);

    JLabel lblPin = new JLabel("Secure PIN :");
    lblPin.setFont(F_BOLD);
    gbc.gridx = 0; gbc.gridy = 2;
    form.add(lblPin, gbc);

    pwdLoginPin = new JPasswordField(14);
    pwdLoginPin.setFont(F_FIELD);
    pwdLoginPin.setEchoChar('*'); // PIN never shown on screen
    applyDigitFilter(pwdLoginPin, PIN_LENGTH);
    pwdLoginPin.setToolTipText("Enter your 4-digit PIN");
    gbc.gridx = 1; gbc.gridy = 2;
    form.add(pwdLoginPin, gbc);

    /* ---- Buttons row (FlowLayout) ---- */
    JPanel buttons = new JPanel(new FlowLayout(FlowLayout.CENTER, 14, 4));
    buttons.setBackground(LIGHT);

    btnLogin = makeButton("Login", BLUE);
    btnLogin.addActionListener(this);
    buttons.add(btnLogin);

    btnClearLogin = makeButton("Clear", GREY);
    btnClearLogin.addActionListener(this);

```

```

buttons.add(btnClearLogin);

gbc.gridx = 0; gbc.gridy = 3; gbc.gridwidth = 2;
gbc.anchor = GridBagConstraints.CENTER;
form.add(buttons, gbc);

lblLoginMessage = new JLabel(" ");
lblLoginMessage.setFont(F_LABEL);
lblLoginMessage.setForeground(RED);
gbc.gridy = 4;
form.add(lblLoginMessage, gbc);

root.add(form, BorderLayout.CENTER);

/* ---- Demo credentials (SOUTH) ---- */
JPanel hint = new JPanel(new GridLayout(2, 1));
hint.setBackground(new Color(232, 238, 248));
hint.setBorder(new EmptyBorder(10, 14, 10, 14));

JLabel h1 = new JLabel("Demo Accounts : 100001/1234 (Arun Kumar) | 100002/2345 (Priya
Sharma)",
    SwingConstants.CENTER);
JLabel h2 = new JLabel("100003/3456 (Rahul Verma) | 100004/4567 (Sneha Nair)",
    SwingConstants.CENTER);
h1.setFont(new Font("Segoe UI", Font.PLAIN, 12));
h2.setFont(new Font("Segoe UI", Font.PLAIN, 12));
h1.setForeground(NAVY);
h2.setForeground(NAVY);
hint.add(h1);
hint.add(h2);

root.add(hint, BorderLayout.SOUTH);

// Pressing ENTER inside either field triggers Login
txtLoginAccount.addActionListener(this);
pwdLoginPin.addActionListener(this);

return root;
}

/* ===== SECTION 7 : DASHBOARD SCREEN (BorderLayout + GridBag + Grid + Flow) ===== */
private JPanel buildDashboardPanel() {
    JPanel root = new JPanel(new BorderLayout(10, 10));
    root.setBackground(LIGHT);
    root.setBorder(new EmptyBorder(0, 0, 0, 0));

    /* ----- HEADER (NORTH) ----- */
    JPanel header = new JPanel(new BorderLayout());
    header.setBackground(NAVY);
    header.setBorder(new EmptyBorder(14, 18, 14, 18));

    lblWelcome = new JLabel("Welcome");
    lblWelcome.setFont(new Font("Segoe UI", Font.BOLD, 18));
    lblWelcome.setForeground(Color.WHITE);

    lblHeaderBalance = new JLabel("Available Balance : Rs. 0.00");
    lblHeaderBalance.setFont(F_BOLD);
    lblHeaderBalance.setForeground(new Color(150, 230, 180));

    lblClock = new JLabel();
    lblClock.setFont(new Font("Segoe UI", Font.PLAIN, 12));
    lblClock.setForeground(new Color(180, 200, 230));

    JPanel right = new JPanel(new GridLayout(2, 1));
    right.setBackground(NAVY);
    lblHeaderBalance.setHorizontalAlignment(SwingConstants.RIGHT);
    lblClock.setHorizontalAlignment(SwingConstants.RIGHT);
    right.add(lblHeaderBalance);
    right.add(lblClock);

    header.add(lblWelcome, BorderLayout.WEST);
    header.add(right, BorderLayout.EAST);
    root.add(header, BorderLayout.NORTH);

    /* ----- CENTRE : transfer form + history table ----- */
    JPanel centre = new JPanel(new BorderLayout(10, 10));
    centre.setBackground(LIGHT);
    centre.setBorder(new EmptyBorder(12, 16, 6, 16));

```

```

centre.add(buildTransferForm(), BorderLayout.NORTH);
centre.add(buildHistoryTable(), BorderLayout.CENTER);
root.add(centre, BorderLayout.CENTER);

/* ----- SOUTH : action buttons + status bar ----- */
JPanel south = new JPanel(new BorderLayout());
south.setBackground(LIGHT);

JPanel actions = new JPanel(new GridLayout(1, 5, 8, 0)); // equal-width buttons
actions.setBackground(LIGHT);
actions.setBorder(new EmptyBorder(6, 16, 8, 16));

btnBalance = makeButton("Balance Inquiry", BLUE);
btnTransfer = makeButton("Fund Transfer", GREEN);
btnHistory = makeButton("Transaction History", NAVY);
btnClearForm = makeButton("Clear Form", GREY);
btnLogout = makeButton("Logout", RED);

btnBalance.addActionListener(this);
btnTransfer.addActionListener(this);
btnHistory.addActionListener(this);
btnClearForm.addActionListener(this);
btnLogout.addActionListener(this);

// Slightly smaller font so long labels never get truncated
for (JButton b : new JButton[]{btnBalance, btnTransfer, btnHistory, btnClearForm, btnLogout}) {
    b.setFont(new Font("Segoe UI", Font.BOLD, 12));
    b.setPreferredSize(new Dimension(140, 38));
}

actions.add(btnBalance);
actions.add(btnTransfer);
actions.add(btnHistory);
actions.add(btnClearForm);
actions.add(btnLogout);

JPanel statusBar = new JPanel(new FlowLayout(FlowLayout.LEFT, 16, 4));
statusBar.setBackground(new Color(232, 238, 248));
lblStatus = new JLabel("Ready.");
lblStatus.setFont(new Font("Segoe UI", Font.PLAIN, 12));
lblStatus.setForeground(NAVY);
statusBar.add(lblStatus);

south.add(actions, BorderLayout.CENTER);
south.add(statusBar, BorderLayout.SOUTH);
root.add(south, BorderLayout.SOUTH);

return root;
}

/** Fund-transfer input form - GridBagLayout keeps labels and fields aligned. */
private JPanel buildTransferForm() {
    JPanel panel = new JPanel(new GridBagLayout());
    panel.setBackground(Color.WHITE);
    panel.setBorder(BorderFactory.createTitledBorder(
        BorderFactory.createLineBorder(new Color(200, 210, 225)),
        " Fund Transfer ", TitledBorder.LEFT, TitledBorder.TOP, F_BOLD, NAVY));

    GridBagConstraints gbc = new GridBagConstraints();
    gbc.insets = new Insets(8, 12, 8, 12);
    gbc.anchor = GridBagConstraints.WEST;
    gbc.fill = GridBagConstraints.HORIZONTAL;

    JLabel l1 = new JLabel("Beneficiary Account :");
    l1.setFont(F_BOLD);
    gbc.gridx = 0; gbc.gridy = 0; gbc.weightx = 0;
    panel.add(l1, gbc);

    cmbToAccount = new JComboBox<>();
    cmbToAccount.setFont(F_LABEL);
    gbc.gridx = 1; gbc.gridy = 0; gbc.weightx = 1.4;
    panel.add(cmbToAccount, gbc);

    JLabel l2 = new JLabel("Amount (Rs.) :");
    l2.setFont(F_BOLD);
    gbc.gridx = 2; gbc.gridy = 0; gbc.weightx = 0;
    panel.add(l2, gbc);

    txtAmount = new JTextField(10);
    txtAmount.setFont(F_FIELD);
    txtAmount.setToolTipText("Positive amount, maximum Rs. " + MAX_TXN_AMOUNT + " per transaction");
}

```

```

gbc.gridx = 3; gbc.gridy = 0; gbc.weightx = 1.0;
gbc.ipadx = 30;
panel.add(txtAmount, gbc);
gbc.ipadx = 0;

JLabel l3 = new JLabel("Confirm PIN :");
l3.setFont(F_BOLD);
gbc.gridx = 4; gbc.gridy = 0; gbc.weightx = 0;
panel.add(l3, gbc);

pwdTxnPin = new JPasswordField(8);
pwdTxnPin.setFont(F_FIELD);
pwdTxnPin.setEchoChar('*');
applyDigitFilter(pwdTxnPin, PIN_LENGTH);
pwdTxnPin.setToolTipText("Re-enter your 4-digit PIN to authorise this transfer");
gbc.gridx = 5; gbc.gridy = 0; gbc.weightx = 0.8;
gbc.ipadx = 20;
panel.add(pwdTxnPin, gbc);
gbc.ipadx = 0;

JLabel note = new JLabel("Note : minimum balance of Rs. " + String.format("%.2f", MIN_BALANCE)
    + " must remain in the account after every transfer.");
note.setFont(new Font("Segoe UI", Font.ITALIC, 12));
note.setForeground(GREY);
gbc.gridx = 0; gbc.gridy = 1; gbc.gridwidth = 6;
panel.add(note, gbc);

return panel;
}

/** Transaction-history table wrapped in a scroll pane. */
private JScrollPane buildHistoryTable() {
    String[] columns = {"Txn ID", "Date & Time", "Type", "Counter Account",
        "Amount (Rs.)", "Balance After (Rs.)", "Status"};

    historyModel = new DefaultTableModel(columns, 0) {
        @Override
        public boolean isCellEditable(int row, int column) {
            return false; // history must be read-only
        }
    };

    tblHistory = new JTable(historyModel);
    tblHistory.setFont(new Font("Segoe UI", Font.PLAIN, 13));
    tblHistory.setRowHeight(26);
    tblHistory.setGridColor(new Color(220, 226, 236));
    tblHistory.setSelectionBackground(new Color(210, 226, 248));
    tblHistory.setAutoCreateRowSorter(true); // click a header to sort

    // Sensible starting widths so no column header is cut off
    int[] widths = {90, 165, 90, 135, 115, 145, 85};
    for (int i = 0; i < widths.length; i++) {
        tblHistory.getColumnModel().getColumn(i).setPreferredWidth(widths[i]);
    }

    JTableHeader th = tblHistory.getTableHeader();
    th.setFont(F_BOLD);
    th.setBackground(NAVY);
    th.setForeground(Color.WHITE);
    th.setReorderingAllowed(false);

    // Colour DEBIT rows red and CREDIT rows green for readability
    DefaultTableCellRenderer renderer = new DefaultTableCellRenderer() {
        @Override
        public Component getTableCellRendererComponent(JTable table, Object value,
            boolean isSelected, boolean hasFocus, int row, int column) {
            Component c = super.getTableCellRendererComponent(
                table, value, isSelected, hasFocus, row, column);
            String type = String.valueOf(table.getValueAt(row, 2));
            String status = String.valueOf(table.getValueAt(row, 6));
            if (!isSelected) {
                c.setBackground(row % 2 == 0 ? Color.WHITE : new Color(248, 250, 253));
                if ("FAILED".equals(status)) c.setForeground(RED);
                else if ("DEBIT".equals(type)) c.setForeground(new Color(150, 40, 40));
                else if ("CREDIT".equals(type)) c.setForeground(GREEN);
                else c.setForeground(Color.DARK_GRAY);
            }
            setHorizontalAlignment(column >= 4 && column <= 5 ? SwingConstants.RIGHT
                : SwingConstants.LEFT);

            return c;
        }
    };
}

```

```

    };
    for (int i = 0; i < tblHistory.getColumnCount(); i++) {
        tblHistory.getColumnModel().getColumn(i).setCellRenderer(renderer);
    }

    JScrollPane scroll = new JScrollPane(tblHistory);
    scroll.setBorder(BorderFactory.createTitledBorder(
        BorderFactory.createLineBorder(new Color(200, 210, 225)),
        " Transaction History ", TitledBorder.LEFT, TitledBorder.TOP, F_BOLD, NAVY));
    scroll.getViewport().setBackground(Color.WHITE);
    return scroll;
}

/* =====
 * SECTION 8 : EVENT HANDLING - single ActionListener for all buttons
 * ===== */
@Override
public void actionPerformed(ActionEvent e) {
    Object src = e.getSource();

    if (src == btnLogin || src == txtLoginAccount || src == pwdLoginPin) {
        handleLogin();
    } else if (src == btnClearLogin) {
        clearLoginFields();
        lblLoginMessage.setText(" ");
    } else if (src == btnBalance) {
        handleBalanceInquiry();
    } else if (src == btnTransfer) {
        handleFundTransfer();
    } else if (src == btnHistory) {
        handleTransactionHistory();
    } else if (src == btnClearForm) {
        clearTransferForm();
        lblStatus.setText("Transfer form cleared.");
    } else if (src == btnLogout) {
        handleLogout();
    }
}

/* -----
 * 8.1 LOGIN
 * ----- */
private void handleLogin() {
    String accNo = txtLoginAccount.getText().trim();
    char[] pin = pwdLoginPin.getPassword(); // char[] , not String

    try {
        /* --- Validation 1 : empty fields --- */
        if (accNo.isEmpty() || pin.length == 0) {
            JOptionPane.showMessageDialog(this,
                "Account Number and PIN cannot be empty.",
                "Input Required", JOptionPane.WARNING_MESSAGE);
            return;
        }
        /* --- Validation 2 : account number format --- */
        if (!accNo.matches("\\d{" + ACC_NO_LENGTH + "}")) {
            JOptionPane.showMessageDialog(this,
                "Invalid account number.\nIt must contain exactly "
                + ACC_NO_LENGTH + " digits (0-9).",
                "Invalid Account Number", JOptionPane.ERROR_MESSAGE);
            return;
        }
        /* --- Validation 3 : PIN format --- */
        if (pin.length != PIN_LENGTH) {
            JOptionPane.showMessageDialog(this,
                "Invalid PIN.\nThe PIN must contain exactly "
                + PIN_LENGTH + " digits.",
                "Invalid PIN", JOptionPane.ERROR_MESSAGE);
            return;
        }
        /* --- Validation 4 : credentials --- */
        Account acc = findAccount(accNo);
        if (acc == null || !pinMatches(pin, acc.getPin())) {
            loginAttempts++;
            int left = MAX_LOGIN_ATTEMPTS - loginAttempts;

            if (left <= 0) {
                JOptionPane.showMessageDialog(this,
                    "Maximum login attempts exceeded.\nThe application will now close "
                    + "for security reasons.",
                    "Account Locked", JOptionPane.ERROR_MESSAGE);
            }
        }
    }
}

```



```

        System.exit(0);
    }
    // Deliberately vague message: never reveal which field was wrong
    lblLoginMessage.setText("Invalid credentials. Attempts remaining : " + left);
    JOptionPane.showMessageDialog(this,
        "Invalid Account Number or PIN.\nAttempts remaining : " + left,
        "Login Failed", JOptionPane.ERROR_MESSAGE);
    pwdLoginPin.setText("");
    return;
}

/* --- Success --- */
currentUser = acc;
loginAttempts = 0;
lblLoginMessage.setText(" ");
loadDashboardForCurrentUser();
cardLayout.show(cardPanel, "DASHBOARD");

JOptionPane.showMessageDialog(this,
    "Login successful.\n\nWelcome, " + acc.getHolderName()
        + "\nAccount Number : " + acc.getAccountNumber(),
    "Login Successful", JOptionPane.INFORMATION_MESSAGE);

} finally {
    Arrays.fill(pin, '\0'); // wipe the PIN from memory immediately
    pwdLoginPin.setText("");
}
}

/** Prepares the dashboard for whoever just logged in. */
private void loadDashboardForCurrentUser() {
    lblWelcome.setText("Welcome, " + currentUser.getHolderName()
        + " (A/C " + currentUser.getAccountNumber() + ")");
    updateHeaderBalance();

    // Fill the beneficiary combo box with every account EXCEPT the user's own
    cmbToAccount.removeAllItems();
    for (Account a : accounts) {
        if (!a.getAccountNumber().equals(currentUser.getAccountNumber())) {
            cmbToAccount.addItem(a);
        }
    }
    clearTransferForm();
    refreshHistoryTable();
    lblStatus.setText("Logged in at " + dateFormat.format(new Date()) + ".");
}

/* -----
 * 8.2 BALANCE INQUIRY
 * ----- */
private void handleBalanceInquiry() {
    if (currentUser == null) return;

    updateHeaderBalance();
    String msg = "Account Holder : " + currentUser.getHolderName()
        + "\nAccount Number : " + currentUser.getAccountNumber()
        + "\nAccount Type : Savings"
        + "\n-----"
        + "\nAvailable Balance : Rs. " + String.format("%.2f", currentUser.getBalance())
        + "\nTransferable : Rs. "
        + String.format("%.2f", Math.max(0, currentUser.getBalance() - MIN_BALANCE))
        + "\n\nEnquiry Time : " + dateFormat.format(new Date());

    JOptionPane.showMessageDialog(this, msg, "Balance Inquiry",
        JOptionPane.INFORMATION_MESSAGE);
    lblStatus.setText("Balance inquiry completed at " + dateFormat.format(new Date()) + ".");
}

/* -----
 * 8.3 FUND TRANSFER - the core transaction logic
 * ----- */
private void handleFundTransfer() {
    if (currentUser == null) return;

    char[] pin = pwdTxnPin.getPassword();
    try {
        /* --- Step 1 : beneficiary selected? --- */
        Account beneficiary = (Account) cmbToAccount.getSelectedItemAt();
        if (beneficiary == null) {
            JOptionPane.showMessageDialog(this,
                "Please select a beneficiary account.",

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        "Beneficiary Required", JOptionPane.WARNING_MESSAGE);
    }
    return;
}
/* --- Step 2 : self-transfer is meaningless --- */
if (beneficiary.getAccountNumber().equals(currentUser.getAccountNumber())) {
    JOptionPane.showMessageDialog(this,
        "You cannot transfer funds to your own account.",
        "Invalid Beneficiary", JOptionPane.ERROR_MESSAGE);
    return;
}
/* --- Step 3 : amount must not be empty --- */
String amountText = txtAmount.getText().trim();
if (amountText.isEmpty()) {
    JOptionPane.showMessageDialog(this,
        "Please enter the amount you wish to transfer.",
        "Amount Required", JOptionPane.WARNING_MESSAGE);
    txtAmount.requestFocus();
    return;
}
/* --- Step 4 : amount format (digits, at most 2 decimals) --- */
if (!amountText.matches("\\d{1,7}(\\.\\d{1,2})?")) {
    JOptionPane.showMessageDialog(this,
        "Invalid amount : \"" + amountText + "\"\n\n"
        + "The amount must be a positive number using digits only,\n"
        + "with at most two decimal places. Example : 1500 or 1500.50",
        "Invalid Amount", JOptionPane.ERROR_MESSAGE);
    txtAmount.selectAll();
    txtAmount.requestFocus();
    return;
}

double amount;
try {
    amount = Double.parseDouble(amountText);
} catch (NumberFormatException ex) { // defensive - should never happen
    JOptionPane.showMessageDialog(this,
        "The amount could not be read as a number.",
        "Invalid Amount", JOptionPane.ERROR_MESSAGE);
    return;
}

/* --- Step 5 : amount must be positive --- */
if (amount <= 0) {
    JOptionPane.showMessageDialog(this,
        "The transfer amount must be greater than zero.",
        "Invalid Amount", JOptionPane.ERROR_MESSAGE);
    return;
}

/* --- Step 6 : per-transaction limit --- */
if (amount > MAX_TXN_AMOUNT) {
    JOptionPane.showMessageDialog(this,
        "Transfer limit exceeded.\n\nMaximum allowed per transaction : Rs. "
        + String.format("%.2f", MAX_TXN_AMOUNT)
        + "\nAmount entered : Rs. " + String.format("%.2f", amount),
        "Limit Exceeded", JOptionPane.WARNING_MESSAGE);
    return;
}

/* --- Step 7 : PIN authorisation --- */
if (pin.length != PIN_LENGTH || !pinMatches(pin, currentUser.getPin())) {
    logTransaction(currentUser.getAccountNumber(), "DEBIT",
        beneficiary.getAccountNumber(), amount,
        currentUser.getBalance(), "FAILED");
    refreshHistoryTable();
    JOptionPane.showMessageDialog(this,
        "Transaction PIN is incorrect.\n\nThe transfer has been declined and "
        + "recorded as a failed attempt.",
        "Authorisation Failed", JOptionPane.ERROR_MESSAGE);
    pwdTxnPin.setText("");
    return;
}

/* --- Step 8 : sufficient balance (including minimum balance rule) --- */
if (amount > currentUser.getBalance() - MIN_BALANCE) {
    logTransaction(currentUser.getAccountNumber(), "DEBIT",
        beneficiary.getAccountNumber(), amount,
        currentUser.getBalance(), "FAILED");
    refreshHistoryTable();
    JOptionPane.showMessageDialog(this,
        "Insufficient balance.\n\nAvailable Balance : Rs. "
        + String.format("%.2f", currentUser.getBalance())
        + "\nMinimum Balance to retain : Rs. " + String.format("%.2f",
MIN_BALANCE)

```

MIN_BALANCE)

```

        + "\nMaximum Transferable : Rs. "
        + String.format("%.2f", Math.max(0, currentUser.getBalance() -
MIN_BALANCE));

        + "\nRequested Amount : Rs. " + String.format("%.2f", amount)
        + "\n\nThe transaction has been cancelled.",
        "Insufficient Balance", JOptionPane.ERROR_MESSAGE);
    return;
}
/* --- Step 9 : final confirmation --- */
int confirm = JOptionPane.showConfirmDialog(this,
    "Please confirm the transfer :\n\n"
    + "From      : " + currentUser.getAccountNumber() + " - "
    + currentUser.getHolderName() + "\n"
    + "To        : " + beneficiary.getAccountNumber() + " - "
    + beneficiary.getHolderName() + "\n"
    + "Amount : Rs. " + String.format("%.2f", amount) + "\n\n"
    + "Proceed with this transaction?",
    "Confirm Fund Transfer", JOptionPane.YES_NO_OPTION,
    JOptionPane.QUESTION_MESSAGE);

if (confirm != JOptionPane.YES_OPTION) {
    lblStatus.setText("Transfer cancelled by the user.");
    return;
}

/* --- Step 10 : commit the transaction --- */
currentUser.debit(amount);
beneficiary.credit(amount);

String txnId = "TXN" + (++txnCounter);
String now   = dateFormat.format(new Date());

// Two records: one for the sender, one for the receiver
transactions.add(new Transaction(currentUser.getAccountNumber(), txnId, now, "DEBIT",
    beneficiary.getAccountNumber(), amount, currentUser.getBalance(), "SUCCESS"));
transactions.add(new Transaction(beneficiary.getAccountNumber(), txnId, now, "CREDIT",
    currentUser.getAccountNumber(), amount, beneficiary.getBalance(), "SUCCESS"));

updateHeaderBalance();
refreshHistoryTable();           // history updates automatically
clearTransferForm();

JOptionPane.showMessageDialog(this,
    "Transfer Successful!\n\n"
    + "Transaction ID : " + txnId + "\n"
    + "Amount Transferred : Rs. " + String.format("%.2f", amount) + "\n"
    + "Beneficiary : " + beneficiary.getAccountNumber() + " - "
    + beneficiary.getHolderName() + "\n"
    + "Date & Time : " + now + "\n"
    + "-----\n"
    + "Updated Balance : Rs. " + String.format("%.2f", currentUser.getBalance()),
    "Transaction Successful", JOptionPane.INFORMATION_MESSAGE);

lblStatus.setText(txnId + " completed successfully at " + now + ".");
} finally {
    Arrays.fill(pin, '\0');           // always wipe the PIN, success or failure
    pwdTxnPin.setText("");
}
}

/* -----
 * 8.4 TRANSACTION HISTORY
 * ----- */
private void handleTransactionHistory() {
    if (currentUser == null) return;

    int rows = refreshHistoryTable();

    double credits = 0, debits = 0;
    for (Transaction t : transactions) {
        if (t.getOwner().equals(currentUser.getAccountNumber())
            && "SUCCESS".equals(t.getStatus())) {
            if ("CREDIT".equals(t.getType())) credits += t.getAmount();
            if ("DEBIT".equals(t.getType())) debits += t.getAmount();
        }
    }

    JOptionPane.showMessageDialog(this,
        "Transaction History for A/C " + currentUser.getAccountNumber() + "\n"
        + "-----\n"

```

```

        + "Total Records Displayed : " + rows + "\n"
        + "Total Credited : Rs. " + String.format("%.2f", credits) + "\n"
        + "Total Debited : Rs. " + String.format("%.2f", debits) + "\n"
        + "Current Balance : Rs. " + String.format("%.2f", currentUser.getBalance()),
        "Transaction History", JOptionPane.INFORMATION_MESSAGE);

    lblStatus.setText(rows + " transaction record(s) loaded into the table.");
}

/** Rebuilds the JTable from the ArrayList; returns the number of rows shown. */
private int refreshHistoryTable() {
    historyModel.setRowCount(0); // clear the table first
    if (currentUser == null) return 0;

    int count = 0;
    for (Transaction t : transactions) {
        if (t.getOwner().equals(currentUser.getAccountNumber())) {
            historyModel.addRow(new Object[]{
                t.getTxnId(),
                t.getDateTime(),
                t.getType(),
                t.getCounterAccount(),
                String.format("%.2f", t.getAmount()),
                String.format("%.2f", t.getBalanceAfter()),
                t.getStatus()
            });
            count++;
        }
    }
    return count;
}

/* -----
 * 8.5 LOGOUT
 * ----- */
private void handleLogout() {
    int confirm = JOptionPane.showConfirmDialog(this,
        "Do you really want to log out of Net Banking?",
        "Confirm Logout", JOptionPane.YES_NO_OPTION, JOptionPane.QUESTION_MESSAGE);

    if (confirm == JOptionPane.YES_OPTION) {
        String name = currentUser.getHolderName();
        currentUser = null; // session destroyed
        historyModel.setRowCount(0); // no data left on screen
        cmbToAccount.removeAllItems();
        clearTransferForm();
        clearLoginFields();
        cardLayout.show(cardPanel, "LOGIN");

        JOptionPane.showMessageDialog(this,
            "You have been logged out successfully.\nThank you for banking with us, "
            + name + ".",
            "Logout", JOptionPane.INFORMATION_MESSAGE);
    }
}

/* =====
 * SECTION 9 : HELPER METHODS
 * ===== */

private Account findAccount(String accNo) {
    for (Account a : accounts) {
        if (a.getAccountNumber().equals(accNo)) return a;
    }
    return null;
}

/**
 * Compares a char[] PIN with the stored PIN.
 * The loop never exits early, so the time taken does not leak
 * how many characters were correct.
 */
private boolean pinMatches(char[] entered, String actual) {
    if (entered.length != actual.length()) return false;
    boolean match = true;
    for (int i = 0; i < entered.length; i++) {
        if (entered[i] != actual.charAt(i)) match = false;
    }
    return match;
}

```

```

private void logTransaction(String owner, String type, String counter,
                           double amount, double balanceAfter, String status) {
    transactions.add(new Transaction(owner, "TXN" + (++txnCounter),
                                     dateFormat.format(new Date()), type, counter, amount, balanceAfter, status));
}

private void updateHeaderBalance() {
    if (currentUser != null) {
        lblHeaderBalance.setText("Available Balance : Rs. "
                                + String.format("%.2f", currentUser.getBalance()));
    }
}

private void clearLoginFields() {
    txtLoginAccount.setText("");
    pwdLoginPin.setText("");
    txtLoginAccount.requestFocus();
}

private void clearTransferForm() {
    txtAmount.setText("");
    pwdTxnPin.setText("");
    if (cmbToAccount.getItemCount() > 0) cmbToAccount.setSelectedIndex(0);
}

/** Attaches the digit-only filter to a text field. */
private void applyDigitFilter(JTextField field, int maxLength) {
    ((PlainDocument) field.getDocument()).setDocumentFilter(new DigitOnlyFilter(maxLength));
}

/** Creates a consistently styled button. */
private JButton makeButton(String text, Color bg) {
    JButton b = new JButton(text);
    b.setFont(F_BOLD);
    b.setForeground(Color.WHITE);
    b.setBackground(bg);
    b.setOpaque(true);
    b.setBorderPainted(false);
    b.setFocusPainted(false);
    b.setPreferredSize(new Dimension(150, 38));
    b.setCursor(new Cursor(Cursor.HAND_CURSOR));
    return b;
}

/**
 * A javax.swing.Timer updates the header clock every second.
 * Because it fires on the Event Dispatch Thread, it also demonstrates
 * that the interface keeps repainting while the user works.
 */
private void startClock() {
    Timer timer = new Timer(1000, e ->
        lblClock.setText("Session Time : " + dateFormat.format(new Date())));
    timer.start();
}

/* =====
 * SECTION 10 : MAIN METHOD
 * ===== */
public static void main(String[] args) {
    try {
        UIManager.setLookAndFeel(UIManager.getSystemLookAndFeelClassName());
    } catch (Exception ex) {
        // If the system look-and-feel is unavailable, the default is used.
    }
    // All Swing objects must be created on the Event Dispatch Thread.
    SwingUtilities.invokeLater(() -> new OnlineBankingApp().setVisible(true));
}
}

```

END OF REPORT